

Performance Testing

Date	14 July 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Eligibility Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser Developer Tools
Type of Testing	Load Testing & Functional Performance Testing
Target Module / API	Crop Recommendation Prediction API (/predict)
Test Environment	Hosted Server (Vercel) with Flask Backend
Test Date	14 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits valid crop prediction request	1	10	Prediction returned successfully within 2 seconds
2	Multiple users send prediction requests simultaneously	10	31	All requests processed without errors
3	Invalid or incomplete input values	5	20	Validation error displayed; no server crash
4	Continuous prediction requests	20	61	Stable performance with consistent response times

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)				
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

Step 4: Observations & Analysis

Key Findings:
 [Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
 [List any performance bottlenecks observed during testing]

Optimization Steps Taken:
 [Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]